OPENER: A Frugal Open-World NER Pipeline for the Legal Domain

Thibault Garel **Faiza Belbachir** **Assia Soukane**
LyRIDS, ECE Engineering School, Paris, 75015, France
garel.thibault14@gmail.com fbelbachir@ece.fr assia.soukane@ece.fr

Abstract

Legal text is hard for named entity recognition. New statutes, parties and courts appear continually, and expert annotation is costly. Open-world NER can assign types that were not fixed at training time. These systems are built on general-domain text, though, and legal uses adapt them corpus by corpus and language by language. We take the opposite route, and apply OPENER unchanged, a frugal pipeline of ready-to-use components. Its entity embedder stays generic, and no part of it is ever fitted to a legal schema. We test its two typing heads on four legal corpora, in English, Portuguese and German. One needs no target label at all, the other fits a small linear probe. We compare them against GLiNER, GNER, OWNER and a 4-bit Qwen, on quality, latency and energy at once. The transfer holds in English, and weakens across languages. The label-free head degrades sharply. The supervised probe is barely affected, and scores best overall (35.9 AMI end-to-end, 61.4 on gold mentions). Both stay one to two orders of magnitude cheaper than the generative baselines. Detection, not typing, is what caps every system here. The English detectors recover as little as 10.2% of the German mentions. On the ones they do find, our supervised head assigns the right type 78.7% of the time on average over the four corpora, against 64.6% for the detector alone. Yet typing all of them correctly would still stop at 70 AMI out of 100. Where labels exist, the supervised head is the practical operating point, and in-language detection is where adaptation should go first.

1 Introduction

Named Entity Recognition (NER) extracts entities such as persons, organisations and dates from raw text. It is a foundational step in most information extraction pipelines. Standard NER systems are closed-set. They are trained once on a fixed inventory of types, and can only ever predict those types. Real-world applications rarely afford such a static inventory, and legal text is an extreme case. Entity inventories there are open and shifting, with new statutes, case citations, parties, courts, contractual instruments and jurisdictions. Documents are long and formulaic, annotation requires expert effort, and extraction errors carry material consequences. Re-annotating and retraining a tagger for every new type or sub-domain is rarely practical.

Open-world NER targets precisely this regime. It recognises and types entities whose categories were not fixed at training time. Notable systems reach it by different routes. They score spans against type names (Zaratiana et al., 2024), tag generatively (Ding et al., 2024), cluster a dedicated entity encoder into dynamically named types (Genest et al., 2025), or prompt instruction-tuned models at far higher inference compute (Zhou et al., 2024). All of them are trained and evaluated on general-domain text. Their legal uses then adapt them per corpus, schema and language, either by fine-tuning a tagger or by prompting a large model anew. Few legal deployments can pay that price for every schema, sub-domain and jurisdiction. We take the opposite route, with one open-world pipeline whose components are never fitted to a legal corpus. We measure it component by component, to locate the bottleneck and show where the next annotation or compute budget should go.

We do this with OPENER (Open Partitioning Embedding for Named Entity Recognition, Garel et al., 2026), a frugal open-world pipeline assembled from ready-to-use pretrained components. A frozen open-world detector proposes spans. A light contrastively fine-tuned, Matryoshka-truncatable embedder then represents them. That embedder is fine-tuned on general-domain entity spans, and never on a legal schema. This is a design choice rather than a shortcut. An open-world space must stay unspecialised, or it collapses into the closed-set taggers we are trying to avoid. The pipeline

Proceedings of the Natural Legal Language Processing Workshop 2026, pages 469–483
October 28, 2026 ©2026 Association for Computational Linguistics

also runs one to two orders of magnitude cheaper in latency and energy than trained-encoder and generative baselines. Legal practice is rarely GPU-rich, and needs exactly that. We evaluate its two operating points. OPENER-ZS is a label-free head, built on label-name prototypes, transductive refinement and detector fusion. OPENER-Sup is a balanced linear probe, fit on the legal target labels. Our contributions are:

- **A frugal pipeline for legal NER.** We assemble an open-world pipeline from pretrained components, with *no* legal-domain training of any of them, and we test both of its typing heads.
- **A legal open-world benchmark.** Four legal NER corpora in three languages, English (SEC/EDGAR filings and Indian court judgements), Portuguese and German (Brazilian and federal court decisions). Eight systems are compared end to end under one protocol, both OPENER heads, the standard open-world baselines (GLiNER-S/M/L, GNER-T5 and OWNER), and a 4-bit Qwen2.5-1.5B.
- **A three-axis, seed-averaged protocol.** We report quality (AMI), latency and energy jointly rather than accuracy alone, and average every OPENER score over three embedder-retraining seeds.
- **An explanation of why the two heads diverge, and of where quality is lost.** OPENER-ZS stays usable on English legal text (up to 49.1 gold AMI), but it does not lead the zero-shot field. It trails GLiNER-M/L and, on average, OWNER. OPENER-Sup is barely dented by the domain shift (61.4 mean gold AMI), and posts the best end-to-end score (35.9). Its balanced probe draws boundaries between closely related legal types that a single label-name match cannot. Splitting each end-to-end score into detection and typing shows the probe typing 78.7% of detected mentions correctly, against the detector's own 64.6% (four-corpus means). Even a head typing all of them correctly would stop at ≈70 AMI rather than 100. Mentions the detector never proposes cannot be recovered downstream. Both heads are then capped by the English detectors' recall, down to 18.7% on German for GLiNER-L and 10.2% for GLiNER-M. In-language *detection*, not typing, is therefore the first place further adaptation should go.

2 Related Work

2.1 Open-world and zero-shot NER

The three open-world systems we compare against differ in backbone as much as in approach. GLiNER (Zaratiana et al., 2024) scores spans against candidate type names with a DeBERTaV3 encoder (He et al., 2023). GNER (Ding et al., 2024) generates them with a sequence-to-sequence model (Raffel et al., 2020). OWNER (Genest et al., 2025) trains a dedicated entity encoder, then clusters mentions into dynamically named types. Typing from type descriptions (Aly et al., 2021) needs no labelled example at all. A complementary line distils or prompts billion-parameter language models for open NER, as in UniversalNER (Zhou et al., 2024), GoLLIE (Sainz et al., 2024) and ChatIE (Wei et al., 2023). Our comparison also probes that compute and energy cost, through an instruction-tuned LLM baseline (Qwen2.5-1.5B in 4-bit) sized to the hardware this paper measures everything on. OPENER differs from all of these by reusing a pretrained embedding rather than training an encoder, which is what keeps it cheap enough for the legal setting we target.

2.2 Legal NLP and legal NER

Legal NLP is an active field, spanning summarisation, judgement prediction, question answering and information extraction, as a recent legal-NLP survey documents (Ariai et al., 2025). It is driven by the specialised vocabulary, the long formulaic documents and the jurisdiction-specific entity inventories of legal text. On the modelling side, domain-adapted language models such as LEGAL-BERT (Chalkidis et al., 2020) and multi-task benchmarks such as LexGLUE (Chalkidis et al., 2022) show that in-domain pretraining and evaluation matter for legal NLU.

The dominant approach to legal NER is to train or adapt a supervised tagger for each corpus, schema and language. LegNER (Karamitsos et al., 2025) domain-adapts BERT with span supervision for English legal NER and anonymisation. A Serbian legal NER study (Kalušev and Brkljač, 2025) fine-tunes a BERT-type model for court rulings and releases a new dataset. The LegalEval shared task (SemEval-2023 Task 6) spurred systems such as a

Bi-LSTM with Flair embeddings for Indian court judgements (Ramesh and Eswara, 2023). Even LLM-based studies remain schema-bound. A financial NER study (Lu and Huo, 2025) prompts GPT-4o, LLaMA-3.1 and Gemini-1.5, and reports systematic failure modes on domain entities. These systems achieve strong in-corpus scores, but each one requires labelled data and (re)training tied to one schema and one language. E-NER (Au et al., 2022) annotates U.S. SEC/EDGAR filings with seven classes, including court, government body and legislation/act. Indian-Legal (Kalamkar et al., 2022) tags Indian judgements with 14 types, from judge and lawyer to precedent. LeNER-Br (Luz de Araujo et al., 2018) covers Brazilian court decisions with dedicated *legislation* and *case-law* tags. German-LER (Leitner et al., 2020) annotates German federal decisions with a fine-grained 19-class schema.

Crucially, the E-NER study (Au et al., 2022) reports that NER models trained on general English degrade sharply on legal text, from 90.5 F1 in-domain down to 61.1 once tested on legal filings. That is precisely the distribution shift we study. Rather than retraining a legal-specific tagger per corpus and language, we build a single open-world pipeline that no legal schema ever touches, and we measure how far it goes on all four of them (Section 4).

2.3 Energy-aware evaluation

Reporting efficiency alongside accuracy is increasingly advocated, from the Green AI agenda (Schwartz et al., 2020) to energy audits of NLP training (Strubell et al., 2019), systematic carbon reporting (Henderson et al., 2020; Patterson et al., 2022) and whole-lifecycle views of AI's footprint (Wu et al., 2022). At inference time specifically, a deployment-cost study (Luccioni et al., 2024) finds general-purpose generative models orders of magnitude more expensive per query than task-specific ones. That is precisely the gap this frugal pipeline targets. We adopt the same three-axis view, on quality, latency and energy, measured with CodeCarbon (Courty et al., 2024) and reported in the energy-times-grid-intensity accounting of Lacoste et al. (2019).

3 Method

OPENER is applied to the legal domain without any of its components being fitted to it. The detector and the entity embedder are ready-to-use checkpoints kept frozen, so no gradient step is ever taken on legal text. Only the typing head touches the legal side. OPENER-ZS consumes no legal label at all, deriving its prototypes from the type names and refining them on the unlabelled test mentions. OPENER-Sup fits a linear classifier on the frozen embeddings of the target's training mentions. It consumes legal labels without updating a single encoder weight. We therefore measure how far a frugal open-world pipeline goes on legal NER *without* being retrained on it, which is the regime a legal deployment faces.

3.1 Overview and notation

The pipeline runs in three stages, and Figure 1 shows them in order. The detector maps a sentence s to a set of candidate mentions $\mathcal{M}(s) = \{m_i\}$. Each mention is then embedded *in context*, and a typing head assigns it a type drawn from the target inventory $\mathcal{T}$,

$$e_i = f_\theta(s, m_i) \in \mathbb{R}^D, \qquad \hat{y}_i \in \mathcal{T}, \qquad (1)$$

where f_θ is the frozen embedder and e_i the D-dimensional vector it produces for mention m_i, read together with its sentence s. We keep the full D=768. The predicted type $\hat{y}_i$ is drawn from $\mathcal{T}$, the legal type inventory of the target corpus, which f_θ never saw during its own training. Stages one and two are identical at both operating points. Only the typing head of stage three changes, and Table 9 in Appendix C states exactly how.

3.2 Preprocessing

Three preparation steps sit between a raw legal corpus and the pipeline. Each one is a deliberate choice rather than a formatting detail.

Corpus normalisation The four corpora ship in different formats (CoNLL columns, character spans, token-level tags). We convert them all to one representation, a sentence paired with $(\text{start}, \text{end}, \text{type})$ triples in *character* offsets. A single evaluation path therefore serves every corpus, and no system benefits from a corpus-specific reader. Sentences with no gold entity are dropped, on every system alike, and each test split is capped at its first 1000 entity-bearing sentences.

Span-in-context encoding The embedder never sees a mention in isolation. A mention spanning characters $[a, b)$ of sentence s is encoded as the

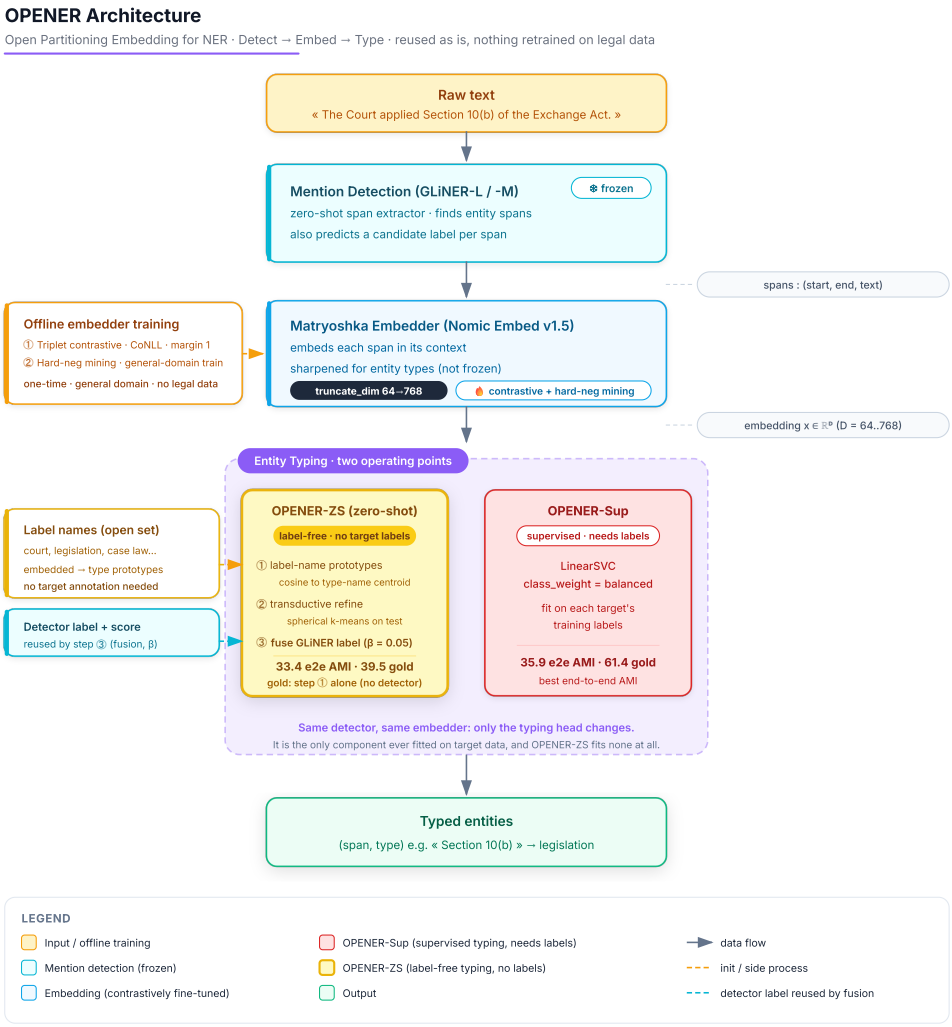


Figure 1: The OPENER pipeline (Detect → Embed → Type), with its two typing heads, OPENER-ZS (label-name prototypes, transductive refinement, detector fusion) and OPENER-Sup (supervised linear probe). The embedder is sharpened once on general-domain CoNLL-2003 spans, never on legal text, and no component is fitted to legal data. AMI figures are three-seed means, end-to-end with the full OPENER-ZS head, on gold with its first step alone (Table 9).

whole sentence with the mention delimited in place,

$$\texttt{classification:}\ s_{<a}\ \texttt{[ENT]}\ s_{a:b}\ \texttt{[/ENT]}\ s_{\geq b},$$

where `classification:` is the literal task prefix Nomic v1.5 expects, colon included. Keeping the sentence lets the same surface form be typed differently in two contexts. Legal text demands that constantly, since a party name can be a company in one clause and a litigant in the next. The markers, rather than a cropped span, keep that context while still telling the encoder which tokens to describe.

Type anchors OPENER-ZS needs a vector per type, and a raw schema code such as `ORGANIZACAO` or `GS` carries little meaning for an English-trained encoder. Each type therefore receives one to five English anchor words, as in `COURT` → *court*, *tribunal*, *judiciary*. They are listed in full in Table 11, and were fixed before any result was seen. End-to-end, every anchor is embedded through the same four templates, *{a}*, *a {a}*, *the {a}* and *an example of {a}*. Each template goes through the span-in-context format above, with the anchor itself marked. The point is distributional. A prototype built this way comes out of the same input format as the mentions it will be compared against. A bare word never appears in that format at inference.

3.3 Mention detection

Spans are proposed by a frozen GLiNER detector (Zaratiana et al., 2024) prompted with the target type names at confidence threshold 0.3, the value released with the model. The two heads run on different sizes, GLiNER-L for OPENER-ZS and GLiNER-M for OPENER-Sup. The detector is a

modular slot, not a tuned hyperparameter. Table 2 reports every GLiNER size on its own, so the effect of that choice stays readable from the baselines rather than hidden inside our pipeline.

3.4 Entity embedding

Mentions are embedded by Nomic v1.5 (Nussbaum et al., 2025), a Sentence-BERT-style (Reimers and Gurevych, 2019), Matryoshka-truncatable embedder (Kusupati et al., 2022). We sharpen it once, with a FaceNet triplet step (Schroff et al., 2015) on English, general-domain CoNLL-2003 spans (Tjong Kim Sang and De Meulder, 2003), followed by an error-driven hard-negative mining pass, the recipe of Garel et al. (2026). Training it away from any legal schema is deliberate. The same single checkpoint then serves all four legal schemas without re-fitting. Nothing in the pipeline is legal-specific, except the type inventory it is asked about, and that is the design this paper puts to the test. We keep the full 768 dimensions, with no Matryoshka truncation. The embedder is not the cost bottleneck here (Table 4), so truncating would trade quality for a saving we do not need. Both embedder checkpoints are public.[1]

3.5 Entity typing

OPENER-ZS The zero-shot head, the open-world default, uses no target labels. Each legal type t is represented by a prototype p_t, the ℓ_2-normalised centroid of the embeddings of the words forming its name,

$$\begin{aligned} \overline{e}_t &= \frac{1}{|\mathrm{name}(t)|} \sum_{w \,\in\, \mathrm{name}(t)} f_\theta(w), \\ p_t &= \overline{e}_t \,/\, \|\overline{e}_t\|_2, \end{aligned} \tag{2}$$

where $\mathrm{name}(t)$ is the set of anchor words naming type t (Section 3.2). Each anchor word w contributes its embedding $f_\theta(w)$, and $\overline{e}_t$ is their average. Dividing that average by its Euclidean norm $\|\overline{e}_t\|_2$ gives a unit-length prototype. End-to-end, $f_\theta(w)$ is replaced by the four-template average of Section 3.2, which changes the vector but not the equation.

A mention is assigned the nearest prototype by cosine similarity. That assignment is then refined by a *transductive* pass of spherical k-means over the unlabelled test mentions, seeded by the prototypes, and *fused* with the detector's own zero-shot label,

$$s_t(i) = \cos(e_i, \tilde{p}_t) + \beta\, g_i\, \mathbb{1}[t = \ell_{\text{GLiNER}}(i)]\,, \tag{3}$$

where $s_t(i)$ is the fused score of assigning type t to mention m_i, and $\tilde{p}_t$ the centroid of type t after that refinement, which is p_t itself when the transductive pass is switched off. The detector contributes the type $\ell_{\text{GLiNER}}(i)$ it predicted for that span and its confidence $g_i \in [0,1]$ in it. The indicator $\mathbb{1}[\cdot]$ is worth 1 only when t is that predicted type. The fusion term therefore only ever raises the score of the detector's own choice. The mention is then typed as $\hat{y}_i = \arg\max_t s_t(i)$. Two constants govern this head, and both are fixed a priori rather than tuned on a legal set. The refinement runs three iterations. That is enough for the centroids to settle on corpora of this size, and few enough to keep them anchored to the label names that seeded them. The fusion weight $\beta{=}0.05$ is small on purpose, so the detector can only break ties the prototypes leave open, instead of overriding them. Appendix D sweeps that bound post-hoc.

OPENER-Sup A balanced linear classifier (scikit-learn's LinearSVC (Pedregosa et al., 2011) with `class_weight=balanced`, built on LIBLINEAR (Fan et al., 2008)) is fit one-vs-rest on the legal target's training labels. For each type c, it minimises a squared-hinge objective with balanced class weights,

$$\begin{aligned} \min_{w_c, b_c}\ & \tfrac{1}{2}\|w_c\|_2^2 \\ & + C \sum_i \omega_{y_i}\, \max\!\big(0,\, 1 - z_{ic}\, h_c(e_i)\big)^2, \end{aligned} \tag{4}$$

with $h_c(e_i) = w_c^\top e_i + b_c$ the one-vs-rest hyperplane of type c, e_i the frozen embedding of training mention i, y_i its gold legal type, $z_{ic} = +1$ if $y_i{=}c$ and -1 otherwise, and C the regularisation strength. The balancing weight $\omega_c = n/(|\mathcal{T}|\, n_c)$ up-weights rare types, with n the total number of training mentions and n_c the count of type c. This matters on legal schemas, where a single type can hold well under 1% of the mentions (Court in E-NER, Table 6). A mention is typed as $\hat{y}_i = \arg\max_c h_c(e_i)$. Rather than matching a single label name, this carves the embedding space into one region per type. It can therefore separate types whose vectors sit close together, as long as their regions do not overlap. This is the supervised upper bound, and it needs target labels. End-to-end, the probe is fit on the detector's own training spans rather

[1] `https://huggingface.co/Thibault-GAREL/opener-sup` and `https://huggingface.co/Thibault-GAREL/opener-zs`.

than on gold ones. It then learns to type the noisy boundaries it will actually be handed.

3.6 Evaluation protocol

We report both *typing-on-gold* (perfect detection) and *end-to-end* protocols. Predicted and gold mentions are aligned by exact character offsets, following the alignment protocol of Genest et al. (2025). Detection errors stay in the evaluation through two sentinel labels, one for a gold mention no system predicted, and one for a predicted span with no gold counterpart. Sentences carrying no gold entity are not evaluated, on any system.

Quality Quality is the Adjusted Mutual Information (AMI) (Vinh et al., 2010) between the gold type partition U and the predicted partition V over the aligned mentions,

$$\mathrm{AMI} = \frac{\mathrm{MI} - \mathbb{E}[\mathrm{MI}]}{\frac{1}{2}[H(U) + H(V)] - \mathbb{E}[\mathrm{MI}]}, \quad (5)$$

where H is the entropy and $\mathrm{MI} = \mathrm{MI}(U, V)$ the mutual information between the two partitions. It measures how much a mention's predicted type reduces the uncertainty about its gold type. Subtracting the chance agreement $\mathbb{E}[\mathrm{MI}]$ makes a random typing score near 0, whatever the number of types, and a perfect one 1. The denominator uses the arithmetic-mean normalisation (the scikit-learn default). We report AMI $\times 100$, so the scale runs from ≈ 0 for chance to 100 for a partition identical to the gold one. The value 100 is reachable in the typing-on-gold protocol, where both partitions cover the same mentions. End-to-end it is not. A missed or spurious span puts a sentinel on one side and a real type on the other, so any detection error caps the score below 100, whatever the typing head does. Section 4.4 quantifies that cap directly.

Why AMI, and what we report beside it AMI is invariant to a relabelling of the predicted types. A system that consistently called *judges* what the gold calls *lawyers*, and conversely, would score 100 AMI and 0 F1. That invariance is the point here. OWNER names its clusters dynamically, and the generative baselines paraphrase type strings, so a metric requiring predicted and gold names to coincide cannot score them at all. AMI is also corrected for chance, which matters across schemas of 6 to 19 unequally frequent types. That argument binds only for systems answering outside the target inventory, though. OPENER's heads, the GLiNER family, GNER and the LLM baseline all answer inside it. For these, an F1 is well defined, and we report it. On gold mentions we give macro-F1 (Table 1). End-to-end we give micro entity-level precision, recall and F1 (Table 3). A span counts as correct only when offsets and type both match, the convention of the four dataset papers. OWNER alone has none.

Cost We also report per-sentence latency (p50) and energy (Wh per 1000 sentences, measured with CodeCarbon (Courty et al., 2024)).

4 Experiments

4.1 Datasets

We evaluate on four legal-specific NER sets across three languages that, together, separate two kinds of distribution shift. *Two are in English*, and isolate a *pure domain shift* from the general-domain spans the embedder was fine-tuned on. They are E-NER (Au et al., 2022) (U.S. SEC/EDGAR filings, 7 types) and Indian-Legal (Kalamkar et al., 2022) (Indian court judgements, 14 fine-grained legal types). LeNER-Br (Luz de Araujo et al., 2018) (Brazilian court decisions) and German-LER (Leitner et al., 2020) (German federal court decisions) add a *cross-lingual* shift. All four annotate *legal-specific* types (courts, statutes, legislation, case law, provisions) rather than the generic person/organisation/date schema of general NER. Table 5 in Appendix A gives their sizes, along with the supervised topline published with each one, from 91.1 to 96.1 entity-level F1. Those figures set the scale our end-to-end numbers should be read against.

4.2 Setup

Embedder, detector and heads are exactly those of Section 3, and nothing is retrained on legal data. The baselines are GLiNER-S/M/L, GNER-T5-base, OWNER and an instruction-tuned LLM baseline, Qwen2.5-1.5B-Instruct in 4-bit, all prompted zero-shot with the target type names. Test splits are capped at the first 1000 entity-bearing sentences, with Qwen on a 200-sentence subsample. That model is the largest our 6 GB GPU runs. A hosted frontier model would have made the latency and energy axes unmeasurable, a constraint we spell out in the Limitations.

Table 12 in Appendix G gives the complete configuration. Every typing hyperparameter of Section 3 is fixed a priori and never tuned on a legal

set, $C{=}1$ for the balanced probe included. Evaluation code, anchor dictionaries, dataset loaders and per-run logs are public.[2]

The two protocols do not exercise the same zero-shot head (Table 9). Table 1 measures the label-name geometry on its own, and Table 2 the operating point one would deploy.

We report two orthogonal sources of uncertainty. The embedder's *training* is stochastic, so the study is repeated under three retraining seeds, every OPENER cell giving mean $\pm$ std, the baselines being frozen checkpoints run once. Inference being deterministic, the remaining uncertainty is the finite test set, quantified with a paired bootstrap in Section 4.3. Efficiency is deliberately *not* seed-averaged, since a seed changes the embedder's weights and not its per-sentence FLOPs.

4.3 Results

Keeping the two protocols apart, Table 1 reports typing-on-gold quality and Table 2 the end-to-end comparison. OWNER appears in both, as the one baseline that types mentions it is given rather than detecting and typing jointly. Three findings stand out.

Label-free typing is strong in English but degrades cross-lingually. OPENER-ZS is usable on the two English sets (47.7 on E-NER and 49.1 on Indian, from English label-name prototypes alone), but it drops to 37.4 (pt) and 23.8 (de). Its English prototypes align well with English legal mentions, far less with other-language ones (Table 1). It also trails OWNER's own label-free typing on all four sets (44.7 mean against 39.5). OWNER trains an entity encoder rather than reusing one, so the label-free operating point is where reuse costs the most. It is also the *seed-sensitive* head. Its gold scores vary by $\pm$3–6 AMI across the three embedder retrainings, where the supervised probe varies by at most $\pm$1.4. The released seed is in fact a low draw on the English sets. Label-name prototypes depend on the fine geometry of the embedding space, which shifts from one contrastive retraining to the next. A fitted probe absorbs those shifts.

Supervised typing transfers well, including across languages, because the separation it learns is geometric, not just statistical. Fitting a balanced linear probe on the same frozen, general-domain embedder reaches 61.4 mean typing-on-gold AMI over the four sets. That is 70.9 and 60.8 on the two English corpora, and still 53.7 and 60.3 on Portuguese and German. The spread across seeds is at most $\pm$1.4 AMI. It leads OWNER on every set, by 16 AMI on E-NER and by 35 on German. That is the same range as its 62.3 mean over 13 general-domain corpora (Garel et al., 2026), although corpora and type counts differ too much for the gap to be read as a cost. A balanced probe fits hyperplanes precise enough to pull apart legal types whose vectors sit *close together*, a lawyer from a judge, or a statute from a case citation. A single label-name match cannot draw that boundary without supervision. Macro-F1 over the same gold predictions tells the same story (89.1 on E-NER, 74.6 on average, against 46.1 for the label-free head), so no conclusion here hinges on the choice of metric. Those figures type mentions that are given, whereas the published toplines of Section 4.1 include detection, so the two are not two sides of one comparison. What the pair does show is that typing is not where a frozen, general-domain space falls short.

End-to-end, detection sets the ceiling, and the label-free head cannot pull ahead of the detector it fuses with. On these legal-specific schemas, every system clusters in a narrow $\approx$ 33–36 AMI band (Table 2). The English detectors recover only a fraction of gold mentions, GLiNER-L between 18.7 and 57.4%, GLiNER-M between 10.2 and 61.1%. With so few mentions detected, no typing head can pull far ahead. Per-dataset winners are specific, OWNER (Genest et al., 2025) on E-NER and GNER-T5 on German, though OWNER loses ten points from E-NER to German, at several times the cost of either OPENER head. Qwen trails throughout, while being the most energy-hungry per sentence. *Averaged over the four sets, OPENER-Sup posts the best end-to-end AMI (*35.9*)*, narrowly ahead of GLiNER-L (35.0) and GLiNER-M (34.9), for every retraining seed. OPENER-ZS (33.4) sits just below OWNER (33.7). The margins are small, as expected when detection is the ceiling. A *paired* bootstrap on the released model ($K{=}1000$) nonetheless makes OPENER-Sup's lead **statistically significant**, at $+1.0$ AMI over both GLiNER-M and GLiNER-L (95% CI $[+0.4, +1.7]$ and $[+0.1, +1.9]$). OPENER-ZS's fusion sits significantly *below* GLiNER-L (-1.3, $[-1.8, -0.8]$), largely deferring to the detector's labels.

[2] https://github.com/Thibault-GAREL/LyRIDS_Law-OPENER

Table 1: Typing-on-gold quality (×100, ↑), in AMI and macro-F1.

System	E-NER (en)	Indian (en)	LeNER-Br (pt)	German (de)	Mean
AMI					
OWNER[§]	54.5	59.0	39.7	25.7	44.7
OPENER-ZS	$47.7_{\pm 3.5}$	$49.1_{\pm 3.0}$	$37.4_{\pm 6.0}$	$23.8_{\pm 4.3}$	$39.5_{\pm 3.0}$
OPENER-Sup[*]	$\mathbf{70.9}_{\pm 1.2}$	$\mathbf{60.8}_{\pm 1.4}$	$\mathbf{53.7}_{\pm 1.2}$	$\mathbf{60.3}_{\pm 1.0}$	$\mathbf{61.4}_{\pm 0.4}$
Macro-F1 (OWNER: n/a[†])					
OPENER-ZS	$59.2_{\pm 7.0}$	$37.6_{\pm 5.3}$	$57.7_{\pm 6.0}$	$29.7_{\pm 5.4}$	$46.1_{\pm 5.3}$
OPENER-Sup[*]	$\mathbf{89.1}_{\pm 0.2}$	$\mathbf{58.3}_{\pm 2.5}$	$\mathbf{74.7}_{\pm 1.5}$	$\mathbf{76.3}_{\pm 1.3}$	$\mathbf{74.6}_{\pm 0.3}$

Typing on gold mentions, detection bypassed. OPENER cells: mean ± std over three embedder-retraining seeds (Section 4.2), test-set bootstrap std ±0.8–1.7 AMI. Best per column in bold, within each block.
[*] Only OPENER-Sup uses target labels. [§] OWNER typing its own clusters, single deterministic run of the released checkpoint, zero-shot transfer as in Table 2.
[†] OWNER names its clusters dynamically, so its predictions carry no label from the target inventory and no F1 can be computed against it. This is the one case the argument of Section 3.6 actually binds, and the reason AMI is our primary axis.

Table 2: Open-world legal NER, end-to-end: quality on each corpus.

Model	E-NER (en)	Indian (en)	LeNER-Br (pt)	German (de)	Avg
AMI (×100, ↑), zero-shot					
GLiNER-S	33.4	33.9	33.7	31.1	33.0
GLiNER-M	33.9	34.8	34.8	36.1	34.9
GLiNER-L	34.7	**35.2**	**36.3**	34.0	**35.0**
GNER-T5	28.7	28.6	33.2	**39.2**	32.4
Qwen-1.5B[‡]	24.9	28.1	24.9	14.5	23.1
OWNER[§]	**38.4**	33.3	34.8	28.4	33.7
OPENER-ZS	$33.4_{\pm 0.2}$	$33.8_{\pm 0.5}$	$33.5_{\pm 1.2}$	$32.8_{\pm 0.4}$	$33.4_{\pm 0.4}$
AMI, with target labels					
OPENER-Sup[*]	$35.7_{\pm 0.3}$	$35.2_{\pm 0.3}$	$34.1_{\pm 0.7}$	$38.5_{\pm 0.3}$	$\mathbf{35.9}_{\pm 0.4}$

Detector mentions, offset/sentinel alignment (Section 3.6). Best per column in bold, within each block, ties included, the two AMI sub-blocks bolded apart since OPENER-Sup is the only row that consumes target labels. OPENER rows are mean ± std over three retraining seeds, baselines are frozen checkpoints with a single deterministic run. Model size and cost in Table 4, per-corpus cost in Table 10.
[*] Only OPENER-Sup uses target labels, every other row is zero-shot. [‡] Qwen AMI on a 200-sentence subsample. [§] OWNER typing its own dynamically named clusters.

4.4 Analysis

Two distribution shifts act together here, and the results separate them. Under a pure *domain* shift (E-NER), the unspecialised embedder already places legal types far enough apart for a label-name prototype to separate them (47.7 gold AMI, with no target label at all). A cross-lingual shift widens the gap, the label-free head dropping to 24–37 while the probe stays high (53.7/60.3).

A stress test on German-LER's fine 19-type schema isolates fineness of separation from language. OPENER-ZS falls to $22.7_{\pm 2.1}$ gold AMI there, as fine German type codes rarely have a faithful English prototype. OPENER-Sup still reaches $61.5_{\pm 0.7}$, *on a par with its coarse score* (60.3) despite nearly three times as many classes. The zero-shot weakness is therefore not the embedding geometry, but the missing supervised signal.

Detection collapses on German, but is that the language or the schema? GLiNER-L recovers 43.9% of the Portuguese mentions, on a par with the 44.5% it reaches on English Indian judgements, against 18.7% on German. Yet German-LER is also the corpus of long citations, since legal-norm and case-law citations make up 62% of its mentions. The boundary analysis below separates the two.

Where end-to-end quality is actually lost. Table 3 splits each end-to-end score into its two stages, the rows of a pair differing *only* in the typing head. The supervised head is then worth more than its +1.0 AMI suggests. On correctly detected mentions, it types 78.7% right, against the detector's own 64.6%. That 14-point gain is a mean over corpora lifted by German. Pooled over mentions it is 81.3 against 75.1. The end-to-end average hides it, because it applies only to the 35.7% of mentions that survive detection. The zero-shot head moves the other way, 64.5% against 70.1%, which is why its fusion sits below the detector it fuses with. Raising that share to 100%, which is what the oracle column reports, lifts the end-to-end score from 35.9 to ≈70 AMI and not to 100, the two being different scales. What remains are the mentions the detector never proposed, which fall into a single sentinel class whatever their gold type, and that bound is itself a generous one (Table 3).

Many detection failures are boundaries, not blindness. Counting a mention as *touched* when a prediction overlaps it without matching both offsets, GLiNER-L touches 68.6% of gold mentions but aligns only 41.1% exactly, and the generative baseline is the extreme case at 77.2% against 39.8% (Table 3). German is the one corpus this does not rescue, GLiNER-L reaching 38.4% there even relaxed. Long spans are not the cause. GLiNER-L touches 94% of Portuguese legislation and 95% of Indian precedents, both long citations, but only 42% of German persons, at 1.2 words on average, against 79 to 100% for persons elsewhere. German fails as a language, not as a schema. An in-language *detector* would therefore lift the ceiling far more than a better typing head.

What the two heads confuse. The confusions are exactly the ones the design predicts (Appendix B). On German-LER, OPENER-ZS maps NORM onto REGULATION 213 times, its dominant error by far, where the probe makes that same confusion 3 times. The two anchor names are near-synonyms in English, while the corpus separates

Table 3: End-to-end quality split into its two stages, four-set means (×100). An arrow marks a head running *on* the detector above it, so the two rows share their detected spans and differ only in the typing head.

System	Detect. (% gold)		Type	End-to-end (×100)		
	exact	touched	acc.†	F1	AMI	oracle
GLiNER-S	34.4	55.1	62.2	22.2	33.0	65.9
GNER-T5	39.8	77.2	56.5	18.4	32.4	74.2
Qwen-1.5B‡	14.2	20.5	(68.2)¶	10.4	23.1	42.2
OWNER§	n/a	n/a	n/a	n/a	33.7	n/a
Same detector, different typing head						
GLiNER-L	41.1	68.6	70.1	25.0	35.0	72.0
↪ OPENER-ZS	41.1	68.6	64.5	23.1	33.7	72.0
GLiNER-M	35.7	63.4	64.6	22.8	34.9	69.8
↪ OPENER-Sup*	35.7	63.4	**78.7**	**25.3**	**35.9**	69.8

† The first two columns are shares of the *gold* mentions, whereas *acc.* is a share of the mentions *already detected*, so their denominators differ. All columns are unweighted means over the four corpora.
¶ Not comparable, from 343 exact detections only, 4 on German (all typed right). Pooled over mentions, 56.9.
exact: gold mentions recovered on both offsets, which is the recall of Section 4.3. *touched*: also counting mentions a prediction overlaps without matching their boundaries, so the gap to *exact* is made of boundary errors. *acc.*: share of the exactly detected mentions given the right type, which isolates the typing head from detection. *F1*: micro entity-level, offsets and type both correct. *oracle*: the ceiling these detected spans allow, each correctly detected span taking its gold type and every spurious span a single shared label. It is a function of the detector alone, hence identical within a pair, and it is a *loose* upper bound, since recognising a span as spurious is detection rather than typing. Every figure here is the released seed alone, so the AMI column can differ slightly from the three-seed means of Table 2 (33.7 against 33.4 for OPENER-ZS, for instance).
* Only OPENER-Sup uses target labels. ‡ Qwen on a 200-sentence subsample per corpus. § OWNER cannot be decomposed here for two independent reasons: it names its clusters dynamically, so neither typing accuracy nor F1 is defined against the target inventory, and it aligns mentions by word index rather than by character offset, so its detection columns would not be commensurable with the rest of the table. Its AMI is repeated from Table 2 for reference.

the types sharply. On Indian-Legal it collapses OTHER-PERSON into procedural roles 123 times, which no label-name prototype can avoid, since those types all name people. Supervision, not a better embedder, is what the label-free head lacks.

Quality against cost. Table 4 closes the picture, with every axis on one row per system. Both heads match the frugal GLiNER family in cost, one to two orders of magnitude below the generative baselines and several times below OWNER. OPENER-Sup does so while leading on both quality measures, at 337M parameters against OWNER's 294M and Qwen's 1.5B. Quality and cost are not traded off against each other here.

Fusion weight β. Sweeping β post-hoc (Appendix D) confirms the reading above. The label-free head interpolates from the refined prototypes alone (32.4 average AMI at β=0) up to the GLiNER-L labels (35.0), reaching a plateau by $\beta\approx0.2$. It can therefore do no better end-to-end than the detector it fuses with. The sweep also shows the transductive pass failing to survive the transfer, which makes the label-free components the ones to re-examine first under a domain and language shift.

Table 4: Quality against cost, four-set means of Table 2 and Table 3.

Model	Params	AMI↑	F1↑	p50 (ms)↓	Energy (Wh)↓
Zero-shot (no target labels)					
GLiNER-S	50M	33.0	22.2	**22**	**0.5**
GLiNER-M	200M	34.9	22.8	37	0.7
GLiNER-L	330M	**35.0**	**25.0**	81	1.6
GNER-T5	220M	32.4	18.4	8289	96.0
Qwen-1.5B‡	1.5B	23.1	10.4	5426	100.5
OWNER§	≈294M	33.7	n/a†	475	14.6
OPENER-ZS	467M	$33.4_{\pm0.4}$	23.1	133	2.7
With target labels					
OPENER-Sup*	337M	$\mathbf{35.9}_{\pm0.4}$	**25.3**	84	1.7

End-to-end. Only AMI is a three-seed mean ± std. F1, latency and energy come from the released seed-42 run alone. Energy per 1000 evaluated sentences (multiply by 0.056 for gCO_2). Best per column in bold, within each sub-block. Params count the whole pipeline, detector plus the 137M embedder, so OPENER-Sup is lighter than OPENER-ZS only because it runs on GLiNER-M rather than GLiNER-L. F1 is the micro entity-level score of Table 3.
* Only OPENER-Sup uses target labels. ‡ Qwen AMI on a 200-sentence subsample, its energy over three sets (Table 10). § OWNER cost is amortised wall-clock. † OWNER names its clusters dynamically, so no F1 is defined against the target inventory (Table 3).

5 Conclusion

We applied OPENER to legal NER, a frugal open-world pipeline of ready-to-use pretrained parts, kept free of any legal training. We evaluated both of its typing heads on four legal corpora in three languages, against the standard open-world baselines.

The two heads diverge sharply on legal text. OPENER-ZS remains a usable label-free option in English, but it does not lead the zero-shot field. Its gold AMI collapses across languages, as its English-centric prototypes stop matching target-language mentions. OPENER-Sup is almost unaffected, holds up even on German-LER's fine 19-type schema, and posts the best end-to-end AMI with a statistically significant lead. The confusion counts show why. The label-free head maps German NORM onto REGULATION 213 times, where the probe does so 3 times. Supervision is therefore the safer operating point wherever labels exist, and OPENER-ZS is best reserved for languages close to the embedder's own.

End-to-end quality remains bounded first by mention *detection*, whichever typing head runs behind it. Typing every detected span correctly would stop at ≈70 AMI, and nearly half of what detection misses are boundary disagreements rather than mentions it never sees. Adapting the detector in-language, rather than the embedder, is the natural next step, and it preserves the frugal decoupled design. Legal NER was the demanding case, and the design holds.

Limitations

Scale of the LLM comparison Our LLM baseline is a 4-bit Qwen2.5-1.5B, and it is the largest instruction-tuned model our hardware runs. It is prompted zero-shot only, so in-context examples might narrow its gap. Every system in this paper is measured on one consumer GPU with 6 GB of VRAM. A 7B model saturates it in 4-bit, and a larger one does not fit at all. Frontier models are reachable only through paid APIs, and this study has no funding line for inference credits. That constraint is not only budgetary. A hosted model would break the very comparison this paper is built on, because two of our three axes cannot be measured through an API. Latency would fold in network round-trips, provider-side batching and queueing, none of which is a property of the model. Energy could not be measured at all, the serving hardware being neither known nor instrumentable by CodeCarbon. Reporting an API model beside locally measured systems would therefore mean dropping the cost axes, or filling them with numbers that are not comparable. We take the honest consequence. We do not know how close a frontier LLM would come to the topline on these corpora. The oracle-typing ceiling of Section 4.4 is our model-independent answer to how much room is left above the systems we could measure.

Test splits are capped at their first 1000 entity-bearing sentences, which affects German-LER only. Sentences carrying no gold entity are excluded, so precision on entity-free text is never measured. OPENER scores are means over three embedder-retraining seeds. The supervised and end-to-end scores are stable (std $\leq$1.4 AMI), but the label-free prototypes vary by up to $\pm$6 AMI across retrainings, so single-seed zero-shot numbers should be read with care. Inference is deterministic for a fixed embedder. Test-set bootstraps on the released model confirm the typing-on-gold scores (std $\pm$0.8–1.7 AMI), and the significance of OPENER-Sup's end-to-end lead. Both hold across resampling seeds 42/7/123. More fundamentally, both the embedder and the detector are English-centric, and used without any legal or language adaptation. That is the design under test here, and also its main exposure. The detectors' low cross-lingual recall, 18.7% on German for GLiNER-L and 10.2% for GLiNER-M, caps end-to-end quality, and is the first thing a follow-up should address.

Ethics Statement

This work runs and evaluates entity recognition on four publicly released legal NER benchmarks, E-NER over SEC/EDGAR filings, an Indian court-judgement set, LeNER-Br and German-LER. They are used only for evaluation, and under their original licenses. Legal documents contain personal data, such as the names of parties, judges and counsel. We introduce no new annotations and release no data. We only run inference over the existing public test splits, and we report no individual-level content, so the study adds no privacy exposure and makes no attempt at re-identification.

Open-world NER is error-prone, and our own results show recall dropping sharply outside English, as low as 10.2% on German. The pipeline is therefore meant as an assistive tool that surfaces candidate entities for human review. It is not meant as a component of automated legal decision-making, where a missed or mistyped entity can carry material consequences. A missed entity is silent, so a human in the loop remains necessary, especially across languages. Finally, a central motivation of this study is energy frugality. We report per-sentence latency and energy alongside accuracy to make the environmental cost of deployment explicit, and to argue for models cheap enough to run on modest hardware.

The authors used Claude Code (Anthropic) to scaffold and debug the evaluation scripts, and to draft and copy-edit this manuscript. The research question, the experimental design, the choice of baselines and datasets, the execution of all experiments and the interpretation of the results are the authors' own. The authors reviewed and edited all content, and take full responsibility for it.

References

Rami Aly, Andreas Vlachos, and Ryan McDonald. 2021. Leveraging type descriptions for zero-shot named entity recognition and classification. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP)*, pages 1516–1528.

Farid Ariai, Joel Mackenzie, and Gianluca Demartini. 2025. Natural language processing for the legal domain: A survey of tasks, datasets, models and challenges. *ACM Computing Surveys*. To appear; arXiv:2410.21306.

Ting Wai Terence Au, Vasileios Lampos, and Ingemar Cox. 2022. E-NER — an annotated named entity

recognition corpus of legal text. In *Proceedings of the Natural Legal Language Processing Workshop 2022 (NLLP)*, pages 246–255. Association for Computational Linguistics.

Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion Androutsopoulos. 2020. LEGAL-BERT: The muppets straight out of law school. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 2898–2904. Association for Computational Linguistics.

Ilias Chalkidis, Abhik Jana, Dirk Hartung, Michael Bommarito, Ion Androutsopoulos, Daniel Martin Katz, and Nikolaos Aletras. 2022. LexGLUE: A benchmark dataset for legal language understanding in English. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4310–4330. Association for Computational Linguistics.

Benoit Courty, Victor Schmidt, Sasha Luccioni, et al. 2024. CodeCarbon: Estimate and track carbon emissions from machine learning computing. Software, v2.4.1.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, pages 10088–10115.

Yuyang Ding, Juntao Li, Pinzheng Wang, Zecheng Tang, Bowen Yan, and Min Zhang. 2024. Rethinking negative instances for generative named entity recognition. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 3461–3475.

Rong-En Fan, Kai-Wei Chang, Cho-Jui Hsieh, Xiang-Rui Wang, and Chih-Jen Lin. 2008. LIBLINEAR: A library for large linear classification. *Journal of Machine Learning Research*, 9:1871–1874.

Thibault Garel, Faiza Belbachir, and Assia Soukane. 2026. OPENER: Open partitioning embedding for named entity recognition. Manuscript under review.

Pierre-Yves Genest, Pierre-Edouard Portier, Előd Egyed-Zsigmond, and Martino Lovisetto. 2025. OWNER: Toward unsupervised open-world named entity recognition. *IEEE Access*, 13:50077–50105.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *The Eleventh International Conference on Learning Representations (ICLR)*.

Peter Henderson, Jieru Hu, Joshua Romoff, Emma Brunskill, Dan Jurafsky, and Joelle Pineau. 2020. Towards the systematic reporting of the energy and carbon footprints of machine learning. *Journal of Machine Learning Research*, 21:1–44.

Prathamesh Kalamkar, Astha Agarwal, Aman Tiwari, Smita Gupta, Saurabh Karn, and Vivek Raghavan. 2022. Named entity recognition in Indian court judgments. In *Proceedings of the Natural Legal Language Processing Workshop 2022 (NLLP)*, pages 184–193. Association for Computational Linguistics.

Vladimir Kalušev and Branko Brkljač. 2025. Named entity recognition for Serbian legal documents: Design, methodology and dataset development. In *Proceedings of the 15th International Conference on Information Society and Technology (ICIST)*. ArXiv:2502.10582.

Ioannis Karamitsos, Nikolaos Roufas, Khalil Al-Hussaeni, and Andreas Kanavos. 2025. LegNER: A domain-adapted transformer for legal named entity recognition and text anonymization. *Frontiers in Artificial Intelligence*, 8:1638971.

Aditya Kusupati, Gantavya Bhatt, Aniket Rege, Matthew Wallingford, Aditya Sinha, Vivek Ramanujan, William Howard-Snyder, Kaifeng Chen, Sham Kakade, Prateek Jain, and Ali Farhadi. 2022. Matryoshka representation learning. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*.

Alexandre Lacoste, Alexandra Luccioni, Victor Schmidt, and Thomas Dandres. 2019. Quantifying the carbon emissions of machine learning. *arXiv preprint arXiv:1910.09700*.

Elena Leitner, Georg Rehm, and Julian Moreno-Schneider. 2020. A dataset of German legal documents for named entity recognition. In *Proceedings of the Twelfth Language Resources and Evaluation Conference (LREC)*, pages 4478–4485. European Language Resources Association (ELRA).

Yi-Te Lu and Yintong Huo. 2025. Financial named entity recognition: How far can LLM go? In *Proceedings of the Joint Workshop of the 9th Financial Technology and Natural Language Processing (FinNLP), the 6th Financial Narrative Processing (FNP), and the 1st Workshop on Large Language Models for Finance and Legal (LLMFinLegal)*, pages 164–168. Association for Computational Linguistics.

Alexandra Sasha Luccioni, Yacine Jernite, and Emma Strubell. 2024. Power hungry processing: Watts driving the cost of AI deployment? In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*.

Pedro Henrique Luz de Araujo, Teófilo Emídio de Campos, Renato R. R. de Oliveira, Matheus Stauffer, Samuel Couto, and Paulo Bermejo. 2018. LeNER-Br: A dataset for named entity recognition in Brazilian legal text. In *Computational Processing of the Portuguese Language (PROPOR)*, volume 11122 of *Lecture Notes in Computer Science*, pages 313–323. Springer.

Zach Nussbaum, John X. Morris, Brandon Duderstadt, and Andriy Mulyar. 2025. Nomic embed: Training

a reproducible long context text embedder. *Transactions on Machine Learning Research (TMLR)*. Published in TMLR 02/2025. OpenReview: https://openreview.net/forum?id=IPmzyQSiQE.

David Patterson, Joseph Gonzalez, Urs Hölzle, Quoc Le, Chen Liang, Lluis-Miquel Munguia, Daniel Rothchild, David So, Maud Texier, and Jeff Dean. 2022. The carbon footprint of machine learning training will plateau, then shrink. *Computer*, 55(7):18–28.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.

Qwen Team. 2025. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21:1–67.

Vinay N. Ramesh and Rohan Eswara. 2023. FlairNLP at SemEval-2023 task 6b: Extraction of legal named entities from legal texts using contextual string embeddings. System description for SemEval-2023 Task 6 (LegalEval). arXiv:2306.02182.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Oscar Sainz, Iker García-Ferrero, Rodrigo Agerri, Oier Lopez de Lacalle, German Rigau, and Eneko Agirre. 2024. GoLLIE: Annotation guidelines improve zero-shot information-extraction. In *The Twelfth International Conference on Learning Representations (ICLR)*.

Florian Schroff, Dmitry Kalenichenko, and James Philbin. 2015. FaceNet: A unified embedding for face recognition and clustering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Roy Schwartz, Jesse Dodge, Noah A. Smith, and Oren Etzioni. 2020. Green AI. *Communications of the ACM*, 63(12):54–63.

Emma Strubell, Ananya Ganesh, and Andrew McCallum. 2019. Energy and policy considerations for deep learning in NLP. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Erik F. Tjong Kim Sang and Fien De Meulder. 2003. Introduction to the CoNLL-2003 shared task: Language-independent named entity recognition. In *Proceedings of the Seventh Conference on Natural Language Learning (CoNLL)*.

Nguyen Xuan Vinh, Julien Epps, and James Bailey. 2010. Information theoretic measures for clusterings comparison: Variants, properties, normalization and correction for chance. *Journal of Machine Learning Research*, 11:2837–2854.

Xiang Wei, Xingyu Cui, Ning Cheng, Xiaobin Wang, Xin Zhang, Shen Huang, Pengjun Xie, Jinan Xu, Yufeng Chen, Meishan Zhang, Yong Jiang, and Wenjuan Han. 2023. ChatIE: Zero-shot information extraction via chatting with ChatGPT. *arXiv preprint arXiv:2302.10205*.

Carole-Jean Wu, Ramya Raghavendra, Udit Gupta, Bilge Acun, Newsha Ardalani, Kiwan Maeng, Gloria Chang, Fiona Aga, Jinshi Huang, Charles Bai, et al. 2022. Sustainable AI: Environmental implications, challenges and opportunities. In *Proceedings of Machine Learning and Systems (MLSys)*.

Urchade Zaratiana, Nadi Tomeh, Pierre Holat, and Thierry Charnois. 2024. GLiNER: Generalist model for named entity recognition using bidirectional transformer. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), Volume 1: Long Papers*, pages 5364–5376. Association for Computational Linguistics.

Wenxuan Zhou, Sheng Zhang, Yu Gu, Muhao Chen, and Hoifung Poon. 2024. UniversalNER: Targeted distillation from large language models for open named entity recognition. In *The Twelfth International Conference on Learning Representations (ICLR)*.

A Datasets

Table 5 gives the size and the published supervised topline of each corpus, and Table 6 the share of test mentions taken by each entity type.

Table 5: Legal NER datasets, with the published supervised topline of each.

Dataset	Lang	Sent.	Ment.	Types	Sup. F1†
E-NER	en	877	1,793	7	96.1
Indian-Legal	en	866	3,028	14	91.1
LeNER-Br	pt	636	1,536	6	92.5
German-LER	de	1,000¶	1,740	7	96.0

Test sentences carrying ≥ 1 entity, and the mentions actually evaluated. Sentences with no gold entity are evaluated by no system. German-LER is shown with its 7-type *coarse* schema (19-type *fine* schema in Section 4.4).

¶ Of 3,018, the only split above our 1000-sentence cap, every score for it being computed on those first 1000.

† Best entity-level F1 reported with the corpus itself (Au et al., 2022; Kalamkar et al., 2022; Luz de Araujo et al., 2018; Leitner et al., 2020), by a tagger trained in-domain that learns detection jointly with typing (cross-validated on E-NER and German-LER, held-out test split on the other two). Not a competitor to a pipeline that retrains nothing, quoted as the scale of the task.

Table 6: Entity types and their share of test mentions.

Dataset	Entity types (share % of test mentions)
E-NER (en)	Business 41.8, Misc 18.3, Legislation/Act 13.5, Location 11.3, Person 7.6, Government 7.1, Court 0.3
Indian-Legal (en)	Lawyer 16.7, Court 9.3, Respondent 8.8, Other-person 8.7, Provision 8.5, Statute 7.3, Date 7.1, Petitioner 6.0, then GPE, Precedent, Org, Judge, Case-number and Witness, from 5.9 down to 1.9
LeNER-Br (pt)	Organização 32.6, Legislação 24.6, Pessoa 15.2, Tempo 12.5, Jurisprudência 12.0, Local 3.1
German-LER (de)	Legal-norm 38.3, Case-law 23.4, Organization 15.0, Regulation 6.7, Person 6.1, Legal-literature 5.9, Location 4.7

German-LER in its 7-type coarse schema. Legal-norm and case-law together account for 61.7% of its mentions, and both are long multi-token citations, which is what makes exact-offset detection hardest on that corpus.

B Error analysis

Table 7 lists, for each corpus, the typing confusions each head makes most often over the mentions its detector recovered exactly, and Table 8 shows real mentions on which at least one head errs. Both are computed on the released seed.

Three patterns run through them. Types whose *names* are near-synonyms defeat the label-free head and not the probe, German NRM against REG being the extreme case (213 against 3). Types that share a surface form and differ only by procedural role, as the Indian person types do, defeat prototypes by construction, since the anchor words cannot encode a role. And the probe is not uniformly better. On LeNER-Br it sends ORGANIZACAO to LOCAL as often as the label-free head does, 15% of the detected organisations for both (41 against 36 in raw counts).

Table 7: Most frequent typing confusions (gold → predicted), on exactly detected mentions.

	OPENER-ZS		OPENER-Sup	
Corpus	confusion	n	confusion	n
E-NER	GOV → BUS	65	BUS → MISC	18
	BUS → MISC	29	MISC → LEG	10
Indian	O-PER → PETIT	66	RESP → PREC	50
	LAWYER → PREC	60	PETIT → PREC	20
	O-PER → WITN	38	PETIT → O-PER	20
LeNER-Br	ORG → LOCAL	36	ORG → LOCAL	41
	TEMPO → LOCAL	22	ORG → LEGIS	15
German	NRM → REG	**213**	NRM → RS	9
	PER → ORG	12	REG → NRM	6

Raw counts, over different denominators. The heads run on different detectors (GLiNER-L for ZS, GLiNER-M for Sup), so they type different numbers of exactly detected mentions, 326 against 177 on German. As a share of the detected gold type, NRM → REG is 96% against 3.5%, and LeNER-Br ORG → LOCAL 15% for both.
Type codes abbreviated. German: NRM legal norm, REG regulation, RS case law. Indian: O-PER other-person, PETIT petitioner, RESP respondent, PREC precedent, WITN witness.

Table 8: Real mentions on which at least one typing head errs.

Mention	Gold	ZS	Sup
E-NER (en)			
NXK, JSN, JGV	BUS	MISC	BUS
Internet	MISC	GOV	MISC
LeNER-Br (pt)			
Superior Tribunal Militar	ORG	JURISP	ORG
União	ORG	LOCAL	ORG
German-LER (de)			
§ 63 Abs. 1 AO	NRM	REG	NRM
§ 130a VwGO	NRM	REG	NRM
35. BImSchV	NRM	REG	REG
Indian-Legal (en)			
Najmi Waziri	JUDGE	PREC	PREC
State Of Gujarat	RESP	GPE	PREC

The opaque E-NER tickers carry no lexical signal a label name could match, so the zero-shot head defaults to MISC while the probe recovers them from the distribution. The last German row and the two Indian rows are cases both heads get wrong. Each row is one occurrence. Typing is contextual (Section 3.2), so a recurring mention can be typed differently across sentences. The zero-shot head types União right in 3 of its 4 detected occurrences, and State Of Gujarat in 2 of 3.

C The two protocols side by side

Table 9: What differs between the two protocols. Stages one and two are shared, only the typing head changes.

	Typing on gold	End-to-end
Mentions typed	gold spans	detector spans
ZS prototypes	bare anchor words	anchors in 4 templates
Transductive pass	no	yes (3 iterations)
Detector fusion	no detector in the loop	yes (β=0.05)
Sup probe fit on	gold train mentions	detected train mentions
Reported in	Table 1	Table 2

Typing on gold isolates the typing head, so the zero-shot side reduces to its first step: with no detector there is no label to fuse, and re-estimating prototypes on a gold mention set would measure something the deployed system never sees.

D Fusion weight sweep

The weight β of Equation (3) balances the label-name centroids against the detector's own labels. It is fixed a priori at 0.05, and sweeping it post-hoc on the released model is a pure inference knob that leaves the embedder untouched. At β=0 the refined prototypes alone reach 32.4 average AMI (30.5 without the transductive pass), and as β grows the head interpolates up to the GLiNER-L labels (35.0), a plateau reached by $\beta\approx$0.2. The label-free head can therefore do no better end-to-end than the detector it fuses with.

The sweep also exposes a component that does not survive the transfer. At β=0.05, dropping the transductive pass would *raise* the average from 33.7 to 34.8 (released seed, hence the offset from the 33.4 of Table 2). Re-estimating prototypes on test mentions helps when those are mostly true entities, as they are in our general-domain runs (Garel

et al., 2026), whereas here two thirds of the detected spans are alignment noise and the centroids drift towards it. We keep the released configuration rather than tune it on legal test sets.

E Cost per corpus

Table 10 breaks the two cost axes down per corpus. Table 4 reports their four-set means, which is what the discussion uses, the per-corpus spread being driven by sentence length rather than by the system.

Table 10: End-to-end latency and energy, per corpus.

Model	E-NER (en)	Indian (en)	LeNER-Br (pt)	German (de)	Avg
Latency, median per sentence (ms, ↓)					
GLiNER-S	**20**	**22**	**24**	**22**	**22**
GLiNER-M	36	35	37	39	37
GLiNER-L	65	88	92	78	81
GNER-T5	2897	15784	8768	5705	8289
Qwen-1.5B‡	4852	8068	5929	2857	5426
OWNER§	328	755	420	398	475
OPENER-ZS	104	140	172	115	133
OPENER-Sup*	78	102	85	73	84
Energy per 1000 evaluated sentences (Wh, ↓)					
GLiNER-S	**0.4**	**0.5**	**0.5**	**0.5**	**0.5**
GLiNER-M	0.7	0.8	0.8	0.7	0.7
GLiNER-L	1.3	1.7	1.8	1.6	1.6
GNER-T5	49.2	161.0	106.3	67.3	96.0
Qwen-1.5B‡	80.5	157.7	(2886)¶	63.3	100.5
OWNER§	10.0	23.1	12.9	12.2	14.6
OPENER-ZS	2.1	3.3	3.4	2.1	2.7
OPENER-Sup*	1.4	2.4	1.6	1.3	1.7

Measured on the released seed-42 run, same machine for every system. For CO_2, count 0.056 gCO_2 per Wh, the 56 gCO_2/kWh grid intensity CodeCarbon applied to every run here. Every Avg is computed on unrounded values, so it can differ by one unit in the last digit from the average of the rounded cells printed here.

* Only OPENER-Sup uses target labels. ‡ Qwen on a 200-sentence subsample. § OWNER cost is amortised wall-clock, not a per-call latency.

¶ A value in parentheses is left out of its row's Avg, which then uses the other three sets. The meter charges wall-clock time, and this run took 18 h for its 200 sentences where its own median latency implies about 20 min, so a long idle stretch of the machine was billed to it as computation. Its latency, measured per sentence, is unaffected and is kept.

F Type anchors

Table 11 gives the full anchor dictionary used by OPENER-ZS. Every entry is English, including for the Portuguese and German schemas, and all of them were fixed before any result was seen. Two entries explain most of the label-free errors of Appendix B. On German, NRM and REG are given near-synonymous English names, which is why the zero-shot head merges them 213 times. On Indian-Legal, the four person types are separated by a procedural role that no anchor word can carry.

Table 11: The English anchor words naming each legal type.

Type	Anchor words
E-NER (en)	
BUSINESS	business, company, corporation, firm
COURT	court, tribunal, judiciary
GOVERNMENT	government, agency, authority, ministry
LEGISLATION/ACT	legislation, act, law, statute
LOCATION	location, place, country, city
PERSON	person, individual, name
MISCELLANEOUS	miscellaneous, other, entity, thing
Indian-Legal (en)	
CASE_NUMBER	case number, docket number, case no
COURT	court, tribunal
DATE	date, day, time
GPE	location, place, state, city, country
JUDGE	judge, justice
LAWYER	lawyer, attorney, counsel, advocate
ORG	organization, company, institution
OTHER_PERSON	person, individual, name
PETITIONER	petitioner, plaintiff, appellant
PRECEDENT	precedent, case law, prior case, judgment
PROVISION	provision, section, clause, article
RESPONDENT	respondent, defendant, appellee
STATUTE	statute, act, law
WITNESS	witness
LeNER-Br (pt)	
PESSOA	person, individual, name
ORGANIZACAO	organization, company, institution
LOCAL	location, place, city, country
TEMPO	time, date, period
LEGISLACAO	legislation, law, statute, act
JURISPRUDENCIA	case law, jurisprudence, court decision, precedent
German-LER (de), coarse schema	
PER	person, individual, name
LOC	location, place, city, country
ORG	organization, company, institution
NRM	legal norm, law, statute
REG	regulation, ordinance, directive
RS	case law, court decision, ruling
LIT	legal literature, commentary, reference

Released with the code as `configs/legal_anchors.yaml`. A type whose name is missing falls back to the words of the schema code itself.

G Full evaluation setup

Table 12: Legal evaluation setup.

Component	Setting
Entity embedder	Nomic v1.5, D=768, span-in-context, released checkpoint loaded as-is, *not* retrained (Nussbaum et al., 2025)
Detector (OPENER-ZS)	GLiNER-L, frozen, threshold 0.3 (Zaratiana et al., 2024)
Detector (OPENER-Sup)	GLiNER-M, frozen, threshold 0.3, probe fit on its own train detections
Zero-shot typing (end-to-end)	English anchor dictionary (1 to 5 words per type), each anchor embedded in 4 short templates and averaged before ℓ_2 normalisation, then 3 transductive spherical k-means iterations and detector fusion β=0.05
Zero-shot typing (on gold)	plain nearest prototype, single-word anchor embeddings, no transductive pass and no fusion (no detector in the loop)
Supervised typing	LinearSVC one-vs-rest, C=1.0, `class_weight=balanced`, target train labels (Fan et al., 2008)
Baselines	GLiNER-S/M/L (Zaratiana et al., 2024), GNER-T5-base (Ding et al., 2024), OWNER (Genest et al., 2025), Qwen2.5-1.5B-Instruct (Qwen Team, 2025) in NF4 4-bit (Dettmers et al., 2023), all zero-shot on the target type names
Test splits	first 1000 entity-bearing sentences of each test split, Qwen on a 200-sentence subsample
German-LER schema	7-type coarse in the tables, 19-type fine as a stress test
Quality	AMI ($\times$100), typing-on-gold and end-to-end kept apart
Embedder seeds	7 / 42 / 123, full retraining per seed (42 released)
Test-set bootstrap	K=1000, mentions resampled on gold (std $\pm$0.8–1.7 AMI), sentences resampled *paired* end-to-end
Latency / energy	p50 per sentence over warmed-up, GPU-synchronised passes, energy from CodeCarbon normalised per 1000 sentences, released embedder. One consumer GPU (NVIDIA GTX 1660 Ti Max-Q, 6 GB VRAM, CUDA 12.1, PyTorch 2.5.1), same machine for every system, $\approx\pm 15\%$ run-to-run (thermal throttling)
OWNER cost	amortised from wall-clock t: energy $= t \times 110$ W, latency $= t/$sentences

Only the supervised typing head sees legal labels, every other component is a ready-to-use checkpoint. Detector variants are a free modular choice, so each OPENER head takes its best one. Baselines detect and type jointly, hence the end-to-end comparison only. Bootstrap CIs agree to ± 0.1 AMI across resampling seeds 42 / 7 / 123.